

# NBR PATENT LANDSCAPE: SYNTHESIS, COMPOSITION, AND PROCESSING OF NITRILE BUTADIENE RUBBER AND RELATED FORMULATIONS

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## 1. ABSTRACT

This report synthesizes a primary patent set centered on nitrile butadiene rubber (NBR) and closely related nitrile-rubber embodiments, with emphasis on preparation, composition control, and downstream processing. Across the selected patents, the target material is defined either as NBR itself, highly saturated nitrile rubber compositions derived from NBR, or NBR-containing blends and functionalized rubber systems where acrylonitrile-butadiene identity remains central. The landscape spans classical emulsion polymerization of NBR, metathesis-based degradation of NBR in organic solvent, and multiple formulation patents that tune NBR performance through silica chemistry, crosslinking packages, and filler architecture.

The polymerization-oriented disclosures show that NBR can be prepared by aqueous emulsion routes using water, acrylonitrile, 1,3-butadiene, emulsifiers, initiators, and chain-transfer agents, with composition control expressed through nitrile content ranges and monomer feed ratios. Other patents focus on post-polymerization modification or processing of NBR, including metathesis degradation in chlorobenzene with 1-hexene and ruthenium catalysts, where the goal is to reduce molecular weight and improve processability before hydrogenation. Several patents are not synthesis routes per se but are highly relevant because they define formulation windows for NBR compounds: silica CaO content, silica pH, crosslinking agents, peroxide cure systems, and laminar inorganic fillers are used to improve compression set, tensile properties, gasoline permeation resistance, or electro-mechanical behavior.

A recurring trend is the use of water in emulsion polymerization, latex handling, coagulation, washing, or flocculation, while solvent-based patents keep the core reaction medium organic even when water appears in workup. Composition trends include acrylonitrile content windows from about 15 wt% to 60 wt% depending on the embodiment, and performance-driven selection of fillers and crosslinking systems to balance oil resistance, elasticity, processability, and heat resistance. Overall, the selected patents collectively show that NBR innovation is driven by both polymerization chemistry and downstream compounding strategies, with water-rich latex processing dominating synthesis disclosures and formulation engineering dominating property optimization.

## 2. METHODOLOGY

### PRIMARY PATENT EVIDENCE

#### Patent 1

- Patent Number: US8957161B2
- Patent Title: Process for reacting chemical compounds in the presence of catalyst systems
- Assignee: Lanxess Deutschland GmbH
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: This patent discloses metathesis degradation of nitrile rubber in an organic solvent using ruthenium-based catalyst systems and added salts to accelerate catalyst activity. The NBR is dissolved in chlorobenzene, reacted with 1-hexene, and then sampled and quenched for GPC analysis; the invention is directed to improving metathesis efficiency for NBR degradation. - Acrylonitrile content: 35% by weight — General Method for the Metathetic Degradation of Nitrile Rubber - Mooney viscosity (ML 1 + 4 @ 100° C.): 34 Mooney units — General Method for the Metathetic Degradation of Nitrile Rubber - Residual moisture content: 1.8% by weight — General Method for the Metathetic Degradation of Nitrile Rubber - Mw: 240 000 g/mol — General Method for the Metathetic Degradation of Nitrile Rubber - Mn: 100 000 g/mol — General Method for the Metathetic Degradation of Nitrile Rubber - PDI: 2.4 — General Method for the Metathetic Degradation of Nitrile Rubber - 293.3 g chlorobenzene — General Method for the Metathetic Degradation of Nitrile Rubber - 40 g NBR — General Method for the Metathetic Degradation of Nitrile Rubber - 0.8 g (2 phr) of 1-hexene — General Method for the Metathetic Degradation of Nitrile Rubber - 20 g of MCB — General Method for the Metathetic Degradation of Nitrile Rubber - 0.3 ml of chlorobenzene — Example 7 - 0.2 ml of CDCl<sub>3</sub> — Example 7 - 0.15 ml of diethyl diallylmalonate — Example 7 - 1 mg of LiBr — Example 7.02 - 1 mg of CsBr — Example 8.02 - 10 mg of Grubbs II catalyst — Example 7 - 10 mg of Grubbs II catalyst — Example 8 - Medium & Water Role: Core reaction medium is chlorobenzene solution; no water-related process step is disclosed in the extracted NBR degradation evidence. - acrylonitrile content: 35% by weight

**Medium & Water Role** - Core reaction medium is chlorobenzene solution; no water-related process step is disclosed in the extracted NBR degradation evidence. - Core reaction medium: Organic solvent solution polymerization / metathesis degradation in chlorobenzene (MCB)

**acrylonitrile content** - 35% by weight

**Relevant Experimental Evidence** - Example 7: LiBr addition accelerated ring-closing metathesis relative to the salt-free control. - Example 8: CsBr also accelerated ring-closing metathesis, but less strongly than LiBr in the shown data. - General method: NBR was dissolved in chlorobenzene and degraded by metathesis in the presence of Ru catalysts and 1-hexene.

**Technical Relevance** This patent is directly relevant because it treats NBR as the substrate and addresses controlled chain scission by metathesis, a route to lower molecular weight nitrile rubber. It also provides explicit NBR composition data, including acrylonitrile content, which is central to the target attribute.

## Patent 2

- Patent Number: EP2027920B1
- Patent Title: Catalysts for metathesis reactions
- Assignee: LANXESS Deutschland GmbH
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: This patent discloses preparation of ruthenium/osmium carbene metathesis catalysts and their use in metathesis reactions, including nitrile rubber degradation. The examples mainly demonstrate catalyst preparation and metathesis performance with model substrates; the NBR relevance is through catalyst design rather than NBR polymerization. - 5.41 g of 9-fluorenone — Fluorenone-tosylhydrazone (A) - 30 mmol 9-fluorenone — Fluorenone-tosylhydrazone (A) - 5.59 g of toluene-4-sulfonic acid hydrazide — Fluorenone-tosylhydrazone (A) - 30 mmol of toluene-4-sulfonic acid hydrazide — Fluorenone-tosylhydrazone (A) - 30 ml of ethanol — Fluorenone-tosylhydrazone (A) - 30 minutes under reflux — Fluorenone-tosylhydrazone (A) - 2.09 g of fluorenone-tosylhydrazone (1) — 9-Diazo fluorene (B) - 6 mmol of fluorenone-tosylhydrazone (1) — 9-Diazo fluorene (B) - 15 ml of dioxane — 9-Diazo fluorene (B) - 2 ml of 50% aqueous NaOH — 9-Diazo fluorene (B) - 85 °C — 9-Diazo fluorene (B) - 1 h — 9-Diazo fluorene (B) - 10 mg of Grubbs II catalyst — Example 7/8 model metathesis - 0.3 ml of chlorobenzene — Example 7 - 0.2 ml of CDCl<sub>3</sub> — Example 7 - 0.15 ml of diethyl diallylmalonate — Example 7 - 1 mg of LiBr — Example 7.02 - 1 mg of CsBr — Example 8.02 - 5 min, 10 min, 15 min, 20 min, 25 min, 30 min, 45 min, 60 min — conversion tables - Medium & Water Role: Water is present in catalyst precursor preparation as 50% aqueous NaOH and in workup/extraction steps, while the metathesis examples themselves are carried out in organic solvent media. - acrylonitrile content: Not disclosed in extracted evidence

**Medium & Water Role** - Water is present in catalyst precursor preparation as 50% aqueous NaOH and in workup/extraction steps, while the metathesis examples themselves are carried out in organic solvent media. - Core reaction medium: Mixed organic/aqueous catalyst-preparation media; metathesis examples in chlorobenzene/CDCl<sub>3</sub>

**acrylonitrile content** - Not disclosed in extracted evidence

**Relevant Experimental Evidence** - Example 7: LiBr accelerated ring-closing metathesis of diethyl diallylmalonate with Grubbs II catalyst. - Example 8: CsBr also accelerated ring-closing metathesis, with conversion data reported over time. - Catalyst synthesis examples: fluorenone-tosylhydrazone and 9-diazofluorene intermediates were prepared in ethanol/dioxane/aqueous NaOH media.

**Technical Relevance** This patent is relevant because it provides catalyst systems used for metathesis reactions of nitrile rubber, which is a key enabling chemistry for NBR degradation. However, it does not disclose NBR composition or synthesis of NBR itself.

### Patent 3

- Patent Number: US8133948B2
- Patent Title: Nitrile rubber composition, cross-linkable nitrile rubber composition and cross ...
- Assignee: Zeon Corporation
- Jurisdiction: US
- Publication Year: 2012
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: This patent discloses crosslinkable nitrile rubber compositions based on highly saturated nitrile rubber, silica, and crosslinking agents. The NBR-related material is not synthesized in the examples; instead, preformed nitrile rubber is compounded with fillers, silane-treated silica, zinc dimethacrylate, and peroxide cure packages to improve heat resistance and compression set. - Iodine value: 100 or less — Abstract/Claims - Acrylonitrile unit content: 36% — Example 1 - Iodine value: 11 — Example 1 - Mooney viscosity (ML1+4, 100° C.): 85 — Example 1 - 15 parts of zinc dimethacrylate — Example 1 - 1.5 parts of 4,4'-di-( $\alpha,\alpha'$ -dimethylbenzyl)diphenylamine — Example 1 - 1.5 parts of zinc salt of 2-mercaptobenzimidazole — Example 1 - 40 parts of silica (1) — Example 1 - 10 parts of zinc oxide — Example 1 - 8 parts of trimellitic acid ester — Example 1 - 8 parts of 40% product of 1,3-bis(t-butyl peroxyisopropyl)benzene — Example 1 - 3.2 parts of pure organic peroxide — Example 1 - 50° C. — Example 1 kneading - 170° C. for 20 minutes — test specimen preparation - 10 MPa — test specimen preparation - 150° C. for 24 hours — heat aging test - 2 parts of vinyl tris( $\beta$ -methoxyethoxy)silane — Comparative Example 2 - 133 m<sup>2</sup>/g — silica (2) specific surface area, Example 2 - 110 m<sup>2</sup>/g — silica (1) specific surface area, Example 1 - 3.8 — pH of Aerosil R972V, Example 2 - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - acrylonitrile content: Not disclosed in extracted evidence

**Medium & Water Role** - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Solid-state compounding / open-roll mixing; no NBR polymerization disclosed

**acrylonitrile content** - Not disclosed in extracted evidence

**Relevant Experimental Evidence** - Example 1: A highly saturated nitrile rubber composition with silane-treated fumed silica and zinc dimethacrylate showed good physical properties and heat resistance. - Example 2: Replacing the silica with calcinated wet silica still produced a crosslinkable nitrile rubber composition with the same general evaluation protocol. - Comparative examples: wet silica or untreated fumed silica reduced heat resistance, showing the importance of silica treatment.

**Technical Relevance** This patent is relevant because it explicitly defines a nitrile-rubber-based composition and gives a clear acrylonitrile content for the NBR-derived rubber. It advances NBR application performance through compounding rather than polymerization, especially for heat resistance and workability.

#### Patent 4

- Patent Number: US11347156B2
- Patent Title: Electro-conductive member for electrophotography, process cartridge for ...
- Assignee: Canon Kabushiki Kaisha
- Jurisdiction: US
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: This patent discloses an electro-conductive member built from a matrix/domain rubber architecture, where one phase is acrylonitrile-butadiene rubber and the other is a different diene rubber such as SBR. The examples focus on compounding and evaluating electro-conductive members rather than synthesizing NBR, but NBR is explicitly used as one of the rubber phases. - Difference of absolute values of SP values: 0.2 to 4.0 (J/cm<sup>3</sup>)<sup>0.5</sup> — Claims -  $\tan \delta_1/\tan \delta_2$ : 0.45 to 2.00 — Claims - Amount of electro-conductive particles: 50 to 150 parts by mass per 100 parts by mass of the second rubber — Claims - 23° C. —  $\tan \delta$  measurement condition - 50% relative humidity —  $\tan \delta$  measurement condition - 80 Hz —  $\tan \delta$  measurement condition - NBR (1) nitrile content: 15% — Example material list - NBR (2) nitrile content: 41.5% — Example material list - NBR (3) nitrile content: 27.5% — Example material list - NBR (4) nitrile content: 18.0% — Example material list - NBR (5) nitrile content: 35% — Example material list - NBR (6) nitrile content: 40.0% — Example material list - SBR (1) styrene content: 45% — Example material list - SBR (2) styrene content: 25% — Example material list - SBR (3) styrene content: 18% — Example material list - SBR (4) styrene content: 15% — Example material list - SBR (5) styrene content: 40% — Example material list - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - acrylonitrile content: 15%, 41.5%, 27.5%, 18.0%, 35%, 40.0%

**Medium & Water Role** - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Rubber compounding / layer formation; no NBR polymerization disclosed

**acrylonitrile content** - 15%, 41.5%, 27.5%, 18.0%, 35%, 40.0%

**Relevant Experimental Evidence** - Example 1: NBR-based domain composition was used in an electro-conductive member architecture. - Example 3: NBR was used as the matrix rubber in a comparative matrix-domain structure. - Table 8: NBR materials with nitrile contents from 15% to 41.5% were used across the rubber set.

**Technical Relevance** This patent is relevant because it explicitly includes NBR as a core rubber phase and provides multiple NBR grades with defined nitrile contents. Although not a synthesis patent, it is useful for understanding how NBR composition is selected in functional rubber architectures.

## Patent 5

- Patent Number: EP2009050B1
- Patent Title: Crosslinkable nitrile rubber composition and crosslinked rubber product
- Assignee: Zeon Corporation
- Jurisdiction: EP
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: This patent discloses crosslinkable nitrile rubber compositions and crosslinked rubber products based on nitrile rubber containing an  $\alpha,\beta$ -ethylenically unsaturated nitrile monomer unit and, in some embodiments, an  $\alpha,\beta$ -ethylenically unsaturated dicarboxylic acid monoester unit. The key process is compounding preformed nitrile rubber with silica of controlled CaO content and polyamine-based crosslinking agents, followed by primary and secondary crosslinking. - Iodine value: 120 or less — Claims - CaO value: 0.5 wt% or more — Claims - Silica content: 5 to 300 parts by weight per 100 parts by weight of nitrile rubber — Claims - Crosslinking agent content: 0.2 to 20 parts by weight per 100 parts by weight of nitrile rubber — Description - 170°C for 20 minutes — Example 1-1 primary crosslink - 10 MPa — Example 1-1 primary crosslink pressure - 170°C for 4 hours — Example 1-1 secondary crosslink - 150°C for 504 hours — Disk set test - 150°C for 168 hours — O-ring set test - 2.3 wt% CaO — Tokusil GU, Example 1-1 - 1.9 wt% CaO — SILTON R2, Example 1-3 - 21.2 wt% CaO — SOLEX CM, Example 1-4 - <0.1 wt% CaO — Carplex #1120, Comparative Example 1-1 - 85 — compound Mooney for Example 1-1 - 89 — compound Mooney for Example 1-2 - 98 — compound Mooney for Example 1-3 - 88 — compound Mooney for Example 1-4 - 113 — compound Mooney for Comparative Example 1-1 - 40 parts of wet silica — Example 1-1 - 1 part of stearic acid — Example 1-1 - 5 parts of plasticizer — Example 1-1 - 1.5 parts of substituted diphenylamine — Example 1-1 - 1.5 parts of 2-mercaptobenzimidazole — Example 1-1 - 2 parts of

3-di-o-tolylguanidine — Example 1-1 - 3.4 parts of hexamethylene diamine carbamate — Example 1-1 - 2 parts of  $\gamma$ -glycidoxypropyltrimethoxysilane — Example 1-2 - Medium & Water Role: Water is used as the emulsion polymerization medium for nitrile rubber production, and later in analytical/workup steps and silica testing; the compounding/crosslinking examples themselves are solid-state mixing. - acrylonitrile content: 15 to 55 wt% (preferred range in the description); 20 to 50 wt% (particularly preferred range); 36% in Example 1-1 feedstock is not explicitly given for this patent's NBR, so not used

**Medium & Water Role** - Water is used as the emulsion polymerization medium for nitrile rubber production, and later in analytical/workup steps and silica testing; the compounding/crosslinking examples themselves are solid-state mixing. - Core reaction medium: Aqueous emulsion polymerization for nitrile rubber production; later compounding and crosslinking of rubber compositions

**acrylonitrile content** - 15 to 55 wt% (preferred range in the description); 20 to 50 wt% (particularly preferred range); 36% in Example 1-1 feedstock is not explicitly given for this patent's NBR, so not used

**Relevant Experimental Evidence** - Example 1-1: Wet silica with 2.3 wt% CaO gave a crosslinkable nitrile rubber composition with compound Mooney 85 and good compression-set performance. - Example 1-2: Addition of  $\gamma$ -glycidoxypropyltrimethoxysilane further improved performance relative to Example 1-1. - Comparative Example 1-1: Very low CaO silica (<0.1 wt%) gave poorer compression set and higher compound Mooney. - Examples 2-1 to 2-5: Nitrile rubbers containing monoester units and silica with controlled pH/CaO showed improved compression set and workability.

**Technical Relevance** This patent is highly relevant because it directly addresses NBR-based crosslinkable compositions and defines composition windows that control mechanical performance. It also includes explicit nitrile-rubber monomer content ranges and a synthesis route for the nitrile rubber feedstock via emulsion polymerization.

## Patent 6

- Patent Number: EP2463335B1
- Patent Title: Nitrile rubber composition, crosslinked nitrile rubber composition, crosslinked ...
- Assignee: Zeon Corporation
- Jurisdiction: EP
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: This patent discloses nitrile rubber compositions and crosslinked rubber products based on nitrile rubber containing an

$\alpha,\beta$ -ethylenically unsaturated nitrile monomer unit, a conjugated diene unit, and optionally a cationic monomer unit. The examples focus on latex-based compounding and flocculation with graphene oxide in a broader patent context, but for the NBR-relevant portion the key disclosure is the composition window and the use of rubber latex processing. - 25 to 60% by weight of  $\alpha,\beta$ -ethylenically unsaturated nitrile monomer unit — Claims - 40 to 75% by weight of conjugated diene monomer unit — Claims - 0 to 30% by weight of cationic monomer unit — Claims - 0.1 to 20% by weight of cationic monomer unit — Description - 5 to 60% by weight of methyl ethyl ketone-insoluble matter — Claims - 1 to 100 parts by weight of laminar inorganic filler per 100 parts by weight of nitrile rubber — Claims - 0.1 to 10 mass % — aqueous solution of oxycellulose fibers/cellulose nanofibers in description - 10 to 30 micrometers — oxycellulose fiber diameter in Example 1 - 1 to 5 mm — oxycellulose fiber length in Example 1 - 1 mass % — cellulose nanofiber dispersion in Example 1 - 3.3 nm — average fiber diameter of cellulose nanofibers in Example 1 - 225 — average aspect ratio of cellulose nanofibers in Example 1 - 40 mass % — H-NBR latex solid content in Example 1 - 50° C. — drying step in Example 1 - 1.5 mm — roll distance in Example 1 - 0.3 mm — open roll distance in Example 1 - 10 to 30° C. — tight-milling temperature in Example 1 - 1.1 — surface velocity ratio of two rolls in Example 1 - 170° C. for 10 minutes — vulcanization step in Example 1 - Medium & Water Role: Water is central to the latex-based processing and to the preparation of aqueous dispersions, washing, and optional coagulation steps; the disclosed rubber composition examples are not solution-polymerization routes. - acrylonitrile content: 25 to 60% by weight

**Medium & Water Role** - Water is central to the latex-based processing and to the preparation of aqueous dispersions, washing, and optional coagulation steps; the disclosed rubber composition examples are not solution-polymerization routes. - Core reaction medium: Aqueous latex processing / emulsion-based rubber compounding

**acrylonitrile content** - 25 to 60% by weight

**Relevant Experimental Evidence** - Example 1: H-NBR latex was mixed with cellulose nanofibers, dried, tight-milled, and vulcanized to form a reinforced rubber composition. - Example 2: Additional triple-roll milling was used before drying, showing a more intensive dispersion route. - Example 10: A coagulation step was inserted before drying, demonstrating an alternative latex-to-rubber conversion route.

**Technical Relevance** This patent is relevant because it defines NBR-family composition windows and uses latex-based processing that is directly related to nitrile rubber manufacturing. The disclosed monomer content ranges are especially useful for understanding target acrylonitrile content in nitrile-rubber embodiments.

## Patent 7

- Patent Number: US10087293B2
- Patent Title: Method for producing rubber composition and rubber composition
- Assignee: Nissin Kogyo Co., Ltd.

- Jurisdiction: US
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: This patent discloses a method for producing a rubber composition by mixing an aqueous solution containing oxycellulose fibers or cellulose nanofibers with rubber latex, drying the first mixture, and tight-milling the dried mixture. The NBR relevance is indirect: one example uses acrylonitrile-butadiene latex as a comparative flocculated rubber system, but the patent is not about NBR synthesis. - 1.25 mass % TEMPO — Example aqueous solution preparation - 12.5 mass % sodium bromide — Example aqueous solution preparation - 28.4 mass % sodium hypochlorite — Example aqueous solution preparation - 20° C. — oxidation reaction temperature - 120 minutes — oxidation reaction time - 10 mass % TEMPO-oxidized oxycellulose fibers — Example aqueous solution preparation - 1 mass % cellulose nanofibers — Example aqueous dispersion - 3.3 nm — average fiber diameter of cellulose nanofibers - 225 — average aspect ratio of cellulose nanofibers - 40 mass % — H-NBR latex solid content - 10,000 rpm — juicer mixer rotational speed - 50° C. — drying oven temperature - 4 days — drying time - 1.5 mm — roll distance - 0.3 mm — open roll distance - 10 to 30° C. — tight-milling temperature - 1.1 — surface velocity ratio of two rolls - 5 times — repeated tight-milling - 8 parts by mass of peroxide — vulcanization step - 170° C. for 10 minutes — compression molding - 0.1 to 10 mass % — aqueous solution cellulose fiber content in description - 0.1 to 60 parts by mass — cellulose fiber content in rubber composition in description - 245 MPa — high-pressure homogenizer pressure in Comparative Example 2 - 50 mass % — cellulose nanofibers in Comparative Example 2 mixture - 20000 sheets — image output in Example 40 - Medium & Water Role: Water is used extensively as the medium for fiber preparation, latex mixing, dispersion, and washing; the rubber processing itself is latex-based rather than solution-based. - acrylonitrile content: Not disclosed in extracted evidence

**Medium & Water Role** - Water is used extensively as the medium for fiber preparation, latex mixing, dispersion, and washing; the rubber processing itself is latex-based rather than solution-based. - Core reaction medium: Aqueous latex mixing / drying / open-roll dispersion

**acrylonitrile content** - Not disclosed in extracted evidence

**Relevant Experimental Evidence** - Example 1: H-NBR latex and cellulose nanofibers were mixed in water, dried, and tight-milled to form a rubber composition sample. - Example 4: Increasing cellulose nanofiber loading improved tensile and tear properties relative to lower-load examples. - Example 10: A coagulation step before drying shortened the drying time while still producing a reinforced rubber composition.

**Technical Relevance** This patent is only indirectly relevant to NBR because it uses hydrogenated nitrile rubber latex rather than conventional NBR. It is included because the extraction evidence contains a nitrile-rubber latex processing route and comparative acrylonitrile-butadiene latex

references, but it does not disclose NBR synthesis.

## Patent 8

- Patent Number: US9376537B2
- Patent Title: Process for preparing completely delaminated graphene oxide/rubber ...
- Assignee: Beijing University Of Chemical Technology
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: This patent discloses a process for preparing graphene oxide/rubber nanocomposites by dispersing graphene oxide in water, pretreating with surfactant, mixing with rubber latex, then flocculating or spray-drying to isolate the composite before vulcanization. The NBR relevance is indirect but explicit in one example where acrylonitrile-butadiene latex is used to make a graphene oxide/acrylonitrile-butadiene rubber nanocomposite. - 10 g graphite oxide — Example 1 - 10 L water — Example 1 - 2 h ultrasonic treatment — Example 1 - 800 W — Example 1 - 1000 Hz — Example 1 - 25° C. — Example 1 - 0.1 wt % solid content — graphene oxide/water sol in Example 1 - 100 g graphene oxide/water sol — Example 1 - 0.625 g butadiene-vinyl pyridine latex — Example 1 - 40 wt % solid content — butadiene-vinyl pyridine latex in Example 1 - 500 g styrene butadiene latex — Example 1 - 20 wt % solid content — styrene butadiene latex in Example 1 - 1% calcium chloride solution — Example 1 - 0.1 phr graphene oxide — Example 1 - 0.25 phr surfactant — Example 1 - 150° C. — vulcanization temperature in Example 1 - 10 g graphite oxide — Example 2 - 1200 g graphene oxide/water sol — Example 2 - 7.5 g butadiene-vinyl pyridine latex — Example 2 - 1.2 phr graphene oxide — Example 2 - 3 phr surfactant — Example 2 - 2 phr graphene oxide — Example 3 - 5 phr surfactant — Example 3 - 4 phr graphene oxide — Example 4 - 10 phr surfactant — Example 4 - 8 phr graphene oxide — Example 5 - 20 phr surfactant — Example 5 - 16 phr graphene oxide — Example 6 - 40 phr surfactant — Example 6 - 24 phr graphene oxide — Example 7 - 60 phr surfactant — Example 7 - 10 wt % solid content — graphene oxide/water sol in Example 10 - 1 g carboxylic styrene butadiene latex — Example 10 - 30 min stirring — Example 10 - 2000 r/min — Example 10 - 10 phr graphene oxide — Example 10 - 0.3 phr surfactant — Example 10 - 1% calcium chloride solution — Example 10 - 10% sulfuric acid solution — Example 10 - 0.5 wt % solid content — graphene oxide/water sol in Example 11 - 42 wt % solid content — natural epoxy latex in Example 11 - 10 phr graphene oxide — Example 11 - 4.2 phr surfactant — Example 11 - 6% calcium sulphate solution — Example 12 - 10 phr graphene oxide — Example 12 - 4 phr surfactant — Example 12 - 1% calcium sulphate solution — Example 13 - 10 phr graphene oxide — Example 13 - 0.4 phr surfactant — Example 13 - 1% calcium chloride solution — Example 14 - 10 phr graphene oxide — Example 14 - 0.2 phr surfactant — Example 14 - 20 g graphite oxide — Example 15 - 100 mL water — Example 15 - 5 h ultrasonic treatment — Example 15 - 2000 W — Example 15 - 20,000 Hz — Example 15 - 0° C. — Example 15 - 5 phr

graphene oxide — Example 15 - 0.5 phr surfactant — Example 15 - 10 g graphite oxide — Example 17 - 5000 g graphene oxide/water sol — Example 17 - 10 g KH550 — Example 17 - 5 phr graphene oxide — Example 17 - 10 phr surfactant — Example 17 - 0.1% calcium chloride solution — Example 18 - 5 phr graphene oxide — Example 18 - 3.8 phr surfactant — Example 18 - 5% sodium sulphate solution — Example 19 - 5 phr graphene oxide — Example 19 - 2 phr surfactant — Example 19 - 6% sodium sulphate solution — Example 20 - 5 phr graphene oxide — Example 20 - 4 phr surfactant — Example 20 - Medium & Water Role: Water is the primary medium for graphene oxide dispersion and latex compounding; later water is also used for flocculation, washing, and dehydration before vulcanization. - acrylonitrile content: Not disclosed in extracted evidence

**Medium & Water Role** - Water is the primary medium for graphene oxide dispersion and latex compounding; later water is also used for flocculation, washing, and dehydration before vulcanization. - Core reaction medium: Aqueous emulsion compounding / flocculation or spray-drying route

**acrylonitrile content** - Not disclosed in extracted evidence

**Relevant Experimental Evidence** - Example 1: Graphene oxide was dispersed in water, compounded with SBR latex, flocculated with calcium chloride, washed, dehydrated, and vulcanized. - Example 9: Higher graphene oxide loading gave a vulcanizate with improved mechanical and air-barrier properties. - Example 14: Acrylonitrile-butadiene latex was used to make a graphene oxide/acrylonitrile-butadiene rubber nanocomposite, directly linking the process to NBR-family rubber. - Example 15: Spray-drying/flocculation conditions were varied to produce graphene oxide/SBR composites with different loadings. - Example 20: A graphene oxide/SBR/butadiene rubber nanocomposite was prepared by emulsion compounding and flocculation.

**Technical Relevance** This patent is relevant because it explicitly demonstrates a graphene oxide/acrylonitrile-butadiene rubber nanocomposite, showing how NBR-family latex can be processed in water-based composite formation. It is not an NBR synthesis patent, but it is useful for understanding latex handling and downstream compounding of acrylonitrile-butadiene rubber systems.

## Patent 9

- Patent Number: US3297792A
- Patent Title: Blends of (1) graft copolymer of vinyl chloride onto chlorinated polyolefin, (2 ...
- Assignee: Monsanto Co
- Jurisdiction: US
- Publication Year: 1967
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: This patent discloses aqueous emulsion polymerization of vinyl chloride onto chlorinated polyolefin backbones, followed by blending with a rubberlike acrylonitrile-diene copolymer and ethylene-vinyl acetate copolymer. The NBR-related content is in the blend component, where butadiene-acrylonitrile copolymers are used as one of the thermoplastic blend ingredients. - 24 percent chlorine content chlorinated polyethylene — Example I - Number average molecular weight: 39,000 — Example I - 0.9 part epoxidized soybean oil type vinyl plasticizer — Example I - 0.2 part sorbitan monolaurate — Example I - 0.15 part hydroxypropyl methyl cellulose — Example I - 20 parts cool water — Example I - 90.5 parts vinyl chloride monomer — Example I - 57 C — Example I - 0.2 part lauroyl peroxide — Example I - 92% of vinyl chloride monomer polymerized — Example I - 50 parts of resin — Example I - 50 parts of butadiene-acrylonitrile copolymer having 50% acrylonitrile — Example I - 5 parts of ethylene-vinyl acetate copolymer having 25% vinyl acetate — Example I - 160 parts water — Example II - 0.30 part hydroxypropyl methyl cellulose — Example II - 0.26 part lauroyl peroxide — Example II - 50 C for 16 hours — Example II - specific viscosity: 0.583 — Example II - 10 parts of butadiene-acrylonitrile copolymer containing 20% acrylonitrile — Example II - 40 parts of ethylene-vinyl acetate copolymer having 35% vinyl acetate — Example II - 2 parts chlorinated polyethylene in 100 parts vinyl chloride monomer — Example III - 20% acrylonitrile — chloroprene-acrylonitrile copolymer in Example III - 55% vinyl acetate — ethylene-vinyl acetate copolymer in Example III - 15 parts chlorinated polyethylene in 10 parts cyclohexane and 100 parts vinyl chloride monomer — Example VI - 160 parts water containing 0.36 part methyl cellulose — Example VI - 0.26 part lauroyl peroxide — Example VI - 50 C for 16 hours — Example VI - specific viscosity: 0.453 — Example VI - 15% chlorine — chlorinated polyethylene in Example VI - 40% acrylonitrile — cyclopentadiene-acrylonitrile copolymer in Example VI - 49% chlorine — chlorinated polyethylene in Example VII - 35% acrylonitrile — 2,3-dimethyl-1,3-butadiene-acrylonitrile copolymer in Example VII - Medium & Water Role: Water is used as the polymerization medium in the graft/suspension process and in resin washing/drying; the NBR-related copolymers appear later as blend components rather than being synthesized in this patent. - acrylonitrile content: 50% by weight; 20%; 20%; 40%; 35%

**Medium & Water Role** - Water is used as the polymerization medium in the graft/suspension process and in resin washing/drying; the NBR-related copolymers appear later as blend components rather than being synthesized in this patent. - Core reaction medium: Aqueous suspension/emulsion polymerization for the graft resin; later blending of rubber components

**acrylonitrile content** - 50% by weight; 20%; 20%; 40%; 35%

**Relevant Experimental Evidence** - Example I: A graft copolymer was prepared in water and blended with a butadiene-acrylonitrile copolymer containing 50% acrylonitrile. - Example II: A homogeneous granular resin was obtained by aqueous suspension polymerization and then blended with a butadiene-acrylonitrile copolymer containing 20% acrylonitrile. - Example VI: A chlorinated polyethylene solution in cyclohexane was dispersed in water and polymerized, then blended with a cyclopentadiene-acrylonitrile copolymer containing 40% acrylonitrile. - Example VII: A similar route used chlorinated polyethylene (49% chlorine) and blended with a 2,3-dimethyl-1,3-butadiene-acrylonitrile copolymer containing 35% acrylonitrile.

**Technical Relevance** This patent is only indirectly relevant to NBR because it uses butadiene-acrylonitrile copolymers as blend components rather than synthesizing NBR itself. It nevertheless provides explicit acrylonitrile contents for several acrylonitrile-diene copolymers and shows how such rubbers are incorporated into multicomponent compositions.

## PATENT COMPARISON TABLE

| Patent      | Applicant                | Priority | Medium & Water Role                                                                                                                   | acrylonitrile content | Key Findings                                                                                    |
|-------------|--------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|
| US8957161B2 | Lanxess Deutschland Gmbh | 2015     | Core reaction medium is chlorobenzene solution; no water-related process step is disclosed in the extracted NBR degradation evidence. | 35% by weight         | Example 7: LiBr addition accelerated ring-closing metathesis relative to the salt-free control. |

|             |                                |      |                                                                                                                                                                                           |                                     |                                                                                                                             |
|-------------|--------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| EP2027920B1 | LANXESS<br>Deutschland<br>GmbH | 2014 | Water is present in catalyst precursor preparation as 50% aqueous NaOH and in workup/extraction steps, while the metathesis examples themselves are carried out in organic solvent media. | Not disclosed in extracted evidence | Example 7:<br>LiBr accelerated ring-closing metathesis of diethyl diallylmalonate with Grubbs II catalyst.                  |
| US8133948B2 | Zeon<br>Corporation            | 2012 | No water-related process step disclosed in the extracted evidence.                                                                                                                        | Not disclosed in extracted evidence | Example 1:<br>A highly saturated nitrile rubber composition with silane-treated fumed silica and zinc dimethacrylate showed |

|              |                        |      |                                                                                                                                                                                                                      |                                                                                                                                                                                         |                                                                                                                          |
|--------------|------------------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| US11347156B2 | Canon Kabushiki Kaisha | 2022 | No water-related process step disclosed in the extracted evidence.                                                                                                                                                   | 15%, 41.5%, 27.5%, 18.0%, 35%, 40.0%                                                                                                                                                    | Example 1: NBR-based domain composition was used in an electro-conductive member architecture.                           |
| EP2009050B1  | Zeon Corporation       | 2011 | Water is used as the emulsion polymerization medium for nitrile rubber production, and later in analytical/workup steps and silica testing; the compounding/crosslinking examples themselves are solid-state mixing. | 15 to 55 wt% (preferred range in the description); 20 to 50 wt% (particularly preferred range); 36% in Example 1-1 feedstock is not explicitly given for this patent's NBR, so not used | Example 1-1: Wet silica with 2.3 wt% CaO gave a crosslinkable nitrile rubber composition with compound Mooney 85 and goo |

|              |                        |      |                                                                                                                                                                                                                          |                                     |                                                                                                                                          |
|--------------|------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| EP2463335B1  | Zeon Corporation       | 2015 | Water is central to the latex-based processing and to the preparation of aqueous dispersions, washing, and optional coagulation steps; the disclosed rubber composition examples are not solution-polymerization routes. | 25 to 60% by weight                 | Example 1: H-NBR latex was mixed with cellulose nanofibers, dried, tight-milled, and vulcanized to form a reinforced rubber composition. |
| US10087293B2 | Nissin Kogyo Co., Ltd. | 2018 | Water is used extensively as the medium for fiber preparation, latex mixing, dispersion, and washing; the rubber processing itself is latex-based rather than solution-based.                                            | Not disclosed in extracted evidence | Example 1: H-NBR latex and cellulose nanofibers were mixed in water, dried, and tight-milled to form a rubber composition.               |

|             |                                           |      |                                                                                                                                                                                                                   |                                     |                                                                                                                          |
|-------------|-------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| US9376537B2 | Beijing University Of Chemical Technology | 2016 | Water is the primary medium for graphene oxide dispersion and latex compounding; later water is also used for flocculation, washing, and dehydration before vulcanization.                                        | Not disclosed in extracted evidence | Example 1: Graphene oxide was dispersed in water, compounded with SBR latex, flocculated with calcium chloride, washed,  |
| US3297792A  | Monsanto Co                               | 1967 | Water is used as the polymerization medium in the graft/suspension process and in resin washing/drying; the NBR-related copolymers appear later as blend components rather than being synthesized in this patent. | 50% by weight; 20%; 20%; 40%; 35%   | Example I: A graft copolymer was prepared in water and blended with a butadiene-acrylonitrile copolymer containing 50% a |

### 3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US8957161B2 uses NBR as a degradation substrate in chlorobenzene solution, whereas EP2009050B1 and EP2463335B1 are latex/emulsion-based nitrile-rubber composition patents with water-centered processing.
- EP2009050B1 explicitly discloses nitrile-rubber monomer content windows of 15 to 55 wt% (preferred) and 20 to 50 wt% (particularly preferred), while US3297792A reports blend-component acrylonitrile contents of 20%, 35%, 40%, and 50% in acrylonitrile-diene copolymers.
- US8957161B2 and EP2027920B1 are metathesis/catalyst patents; only US8957161B2 directly discloses NBR composition data, while EP2027920B1 is catalyst-centric and does not disclose NBR composition values.
- US8133948B2 and EP2009050B1 are compounding-focused: both use silica/crosslinking packages to improve heat resistance or compression set, but only EP2009050B1 includes a direct nitrile-rubber production example.
- US11347156B2 and US3297792A both use NBR-family rubbers as functional components in blends or layered structures rather than as synthesis targets; their acrylonitrile content disclosures are tied to material selection rather than polymerization control.
- US9376537B2 and US10087293B2 are latex-processing patents; both rely on water-rich media and downstream coagulation/washing or drying, but neither is a conventional NBR synthesis disclosure.
- Across the primary set, the most explicit acrylonitrile-content disclosures appear in EP2009050B1, US11347156B2, and US3297792A; the remaining patents either do not disclose NBR composition values or disclose them only indirectly through non-target embodiments.

### 4. CONCLUSION

Across the selected primary patents, NBR innovation is split between true synthesis/production disclosures and downstream performance engineering. EP2009050B1 is the clearest NBR-production patent, giving an aqueous emulsion route and explicit nitrile-monomer content windows, while US8957161B2 addresses NBR degradation in organic solvent and US3297792A shows acrylonitrile-diene copolymers as blend components. The remaining patents mainly optimize NBR-containing formulations through silica chemistry, latex processing, or functional composite architectures. Overall, the set shows that acrylonitrile content is most often controlled at the polymer-design stage, whereas later patents emphasize compounding, filler selection, and process medium management to tune workability, compression set, and barrier or electro-mechanical performance.

## 5. REFERENCES

- US8957161B2 | Process for reacting chemical compounds in the presence of catalyst systems | Lanxess Deutschland GmbH | US | 2015 | <https://patents.google.com/patent/US8957161B2/en>
- EP2027920B1 | Catalysts for metathesis reactions | LANXESS Deutschland GmbH | EP | 2014 | <https://patents.google.com/patent/EP2027920B1/en>
- US8133948B2 | Nitrile rubber composition, cross-linkable nitrile rubber composition and cross ... | Zeon Corporation | US | 2012 | <https://patents.google.com/patent/US8133948B2/en>
- US11347156B2 | Electro-conductive member for electrophotography, process cartridge for ... | Canon Kabushiki Kaisha | US | 2022 | <https://patents.google.com/patent/US11347156B2/en>
- EP2009050B1 | Crosslinkable nitrile rubber composition and crosslinked rubber product | Zeon Corporation | EP | 2011 | <https://patents.google.com/patent/EP2009050B1/en>
- EP2463335B1 | Nitrile rubber composition, crosslinked nitrile rubber composition, crosslinked ... | Zeon Corporation | EP | 2015 | <https://patents.google.com/patent/EP2463335B1/en>
- US10087293B2 | Method for producing rubber composition and rubber composition | Nissin Kogyo Co., Ltd. | US | 2018 | <https://patents.google.com/patent/US10087293B2/en>
- US9376537B2 | Process for preparing completely delaminated graphene oxide/rubber ... | Beijing University Of Chemical Technology | US | 2016 | <https://patents.google.com/patent/US9376537B2/en>
- US3297792A | Blends of (1) graft copolymer of vinyl chloride onto chlorinated polyolefin, (2 ... | Monsanto Co | US | 1967 | <https://patents.google.com/patent/US3297792A/en>